

LECTURE 15

1. Clinical Trials and CRISPR Progress

This section outlines the process by which new treatments are tested and approved, with a focus on the progress of CRISPR-based therapies.

Phases of a Clinical Trial

Clinical trials occur in **four phases**, each with a different purpose¹.

Phase	Focus	Number of Patients	Status	
Phase I	Focus on safety and determining the proper dose ² .	15 to 50 patients ³ .	Determines if the treatment is safe and what dosage is appropriate.	
Phase II	Focus on effectiveness and monitoring side effects ⁴ .	Less than 100 patients ⁵ .	Assesses if the treatment works for the condition.	
Phase III		Compares the new treatment to existing treatment ⁶ .	Hundreds of people ⁷ .	Confirms effectiveness and monitors for adverse reactions in a large group.

Phase IV	The treatment is approved and available ⁸ .	Thousands of people ⁹ .	Long-term effects are observed after the treatment is in general use ¹⁰ .
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CRISPR Clinical Trials Progress - 2025

The graphic illustrates the progress of CRISPR-based clinical trials across various disease areas as of 2025¹¹.

- **Diseases in Trials:** CRISPR is actively being studied for Blood Disorders (**SCD** - Sickle Cell Disease, **TBT** - β -Thalassemia)¹², **Cancers**¹³, Chronic/Recurrent UTI¹⁴, Blood Infections & Sepsis¹⁵, HIV/AIDS¹⁶, Lupus & Autoimmunity¹⁷, and Cardiovascular Diseases¹⁸.
- **Approval Status:** The trials for **Blood Disorders** (SCD and TBT) show the most advanced progress, moving towards or reaching **Approved** status¹⁹¹⁹¹⁹¹⁹.

2. CRISPR Gene Therapy: Casgevy for Sickle Cell Disease (SCD)

This section focuses on the specific genetic basis of SCD and how the first approved CRISPR gene therapy, **Casgevy**, addresses it.

Hemoglobin Switch

The body naturally regulates the types of hemoglobin produced based on developmental stage (Hemoglobin Switch)²⁰.

- **Before Birth:** Hemoglobin primarily contains **γ -globin** (gamma-globin, Fetal Hemoglobin), which has a high oxygen affinity²¹²¹²¹²¹.
- **After Birth:** The production of γ -globin is suppressed and replaced by **β -globin** (beta-globin, Adult Hemoglobin)²²²²²²²².
 - The gene **KLF1** activates **β -globin** and the gene **BCL11A**²³²³²³.

- **BCL11A in turn represses γ -globin expression**²⁴.
- **Sickle Cell Disease (SCD):** SCD is caused by a mutation in the **β -globin gene**²⁵. This leads to the production of diseased hemoglobin, which causes red blood cells to become **sickle-shaped**²⁶.

The Casgevy Treatment Strategy

The goal to cure SCD is to turn the γ -globin production **back ON**²⁷.

- **Natural Process (After Birth):** The **BCL11A** gene, acting as a **regulatory switch**, produces the **BCL11A protein**, which leads to **$\downarrow \gamma$ -globin production**²⁸²⁸²⁸²⁸.
- **Proposed Treatment (Casgevy):** The proposed treatment uses CRISPR to target and turn **OFF** the **BCL11A regulatory switch**²⁹.
 - This stops the production of the **BCL11A protein**³⁰.
 - This **de-represses** the γ -globin gene, leading to **$\uparrow \gamma$ -globin production**³¹³¹³¹³¹.
 - The resulting fetal hemoglobin is functional and replaces the defective β -globin, producing **normal red blood cells**³².

Casgevy Procedure

The procedure is an *ex vivo* (outside the body) gene therapy³³.

1. **Extraction:** Stem cells are **extracted** from the patient's **bone marrow**³⁴.
 2. **Modification:** The stem cells are **genetically modified** in the lab by turning the **BCL11A gene off** using CRISPR³⁵³⁵³⁵³⁵.
 3. **Re-infusion:** The **genetically engineered stem cells** are then given back to the patient³⁶.
 4. **Cure:** The engineered stem cells make healthy fetal hemoglobin and normal red blood cells, effectively treating the disease³⁷.
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3. Innate Immunity and Antimicrobial Peptides (AMPs)

This section details the layers of the immune system and focuses on **Antimicrobial Peptides (AMPs)** as a component of innate immunity.

The Immune System's Layers

The immune system provides **multiple layers of protection** to **prevent** and **eliminate** **Pathogens**³⁸.

System	Characteristic	Components
Innate Immunity	Found in all animals and plants ³⁹³⁹³⁹³⁹ . Rapid response ⁴⁰ . Recognizes traits shared by broad ranges of pathogens ⁴¹ .	Barrier defenses (Skin, Mucous membranes, Secretions) and Internal defenses (Phagocytic cells, Natural killer cells, Antimicrobial proteins ⁴²).
Adaptive Immunity	Found only in vertebrates ⁴³⁴³⁴³⁴³ . Slower response ⁴⁴ . Recognizes traits specific to particular pathogens using a vast array of receptors ⁴⁵ .	Humoral response (Antibodies in body fluids) ⁴⁶ and Cell-mediated response (Cytotoxic cells defend body cells) ⁴⁷ .

Antimicrobial Peptides (AMPs)

AMPs are short protein fragments that are part of the **Innate Immunity**⁴⁸⁴⁸⁴⁸⁴⁸.

- **Source:** AMPs are made when proteins are degraded or are **encoded by genes**⁴⁹.
Bacteria also produce AMPs to eliminate competition⁵⁰.
- **Mechanism:** Upon infection, AMPs **penetrate into the phospholipid membrane** and **disrupt it by interacting with the lipid bilayer** (Barrel-stave, Toroidal-pore, or Carpet model)⁵¹⁵¹⁵¹⁵¹⁵¹⁵¹⁵¹⁵¹. This creates **holes** in the pathogen's membrane, causing it to **lyse**⁵².
- **Alternative Actions:** AMPs can also interact with cellular proteins and nucleic acids inside the pathogen to disrupt their function (e.g., Inhibit cell wall, Inhibit folding, Inhibit enzyme activity)⁵³⁵³⁵³⁵³⁵³⁵³⁵³⁵³⁵³⁵³⁵³⁵³⁵³⁵³⁵³.

Testing AMP Specificity (The Drosophila Experiment)

The slides outline an experiment to test **how specific AMPs are to one type of pathogen**.

1. **Tagging:** To track AMP production, the **AMP gene is fused with the GFP (Green Fluorescent Protein) gene** and expressed in an animal⁵⁴. The resultant AMP::GFP fusion protein will **fluoresce when translated**⁵⁵.
2. **Functional Assay (Flies):** The experiment is done in **Drosophila (fruit flies)** using two bacteria and two AMP genes (**drosomycin** and **defensin**) as readouts⁵⁶⁵⁶⁵⁶⁵⁶.
3. **Mutant Generation: Mutant flies** that cannot naturally produce AMPs are used to control the experiment (functional assay)⁵⁷. These mutants only make the AMP::GFP fusion gene that is expressed⁵⁸.
4. **Testing Specificity and Function:**
 - Flies are infected with one type of pathogen (e.g., the fungus *Neurospora crassa*⁵⁹⁵⁹⁵⁹⁵⁹).
 - **Fluorescence** is checked to see if the AMP is made in response to that pathogen⁶⁰.
 - **Survival of the flies** is used as the **functional assay** to test if the specific AMP produced can kill the pathogen⁶¹.
5. **Graph Interpretation (Infection with *N. crassa*):** The survival curve shows:
 - The **Mutant** flies (blue line, no functional AMPs) have the lowest survival rate, dropping to near 0% by 96 hours⁶²⁶²⁶²⁶².

- The **Wild type** flies (red line, natural immune system) have the highest survival rate (above 75%)⁶³⁶³⁶³⁶³.
- **Mutant + drosomycin** (green line) shows survival very close to Wild type⁶⁴.
- **Mutant + defensin** (purple line) shows poor protection, slightly better than the Mutant but much worse than drosomycin⁶⁵.
- **Conclusion:** This indicates that **drosomycin is highly effective at neutralizing the fungus *Neurospora crassa***, suggesting a degree of specificity for this pathogen⁶⁶.



Elaboration on the AMP Specificity Experiment

The experiment uses **genetic tools** and a **survival assay** in the model organism *Drosophila* (fruit fly) to determine if different AMPs are specialized to fight different types of pathogens²²²².

1. The Experimental Controls and Setup

- **The Goal:** Test the **specificity** and **function** of two AMPs: **drosomycin** and **defensin**³³³³.
- **Genetic Control (Mutant Flies):** The experiment uses flies that are **mutant** and **cannot naturally produce AMPs**⁴. This is a critical negative control, ensuring that any protective effect observed is due *only* to the specific AMP being tested.
- **Tracking Production (AMP::GFP):** The genes for the two AMPs are **fused with the GFP gene** and expressed in the mutant flies⁵.
 - **Fluorescence** is checked to confirm that the specific AMP (**AMP::GFP**) is produced in response to the pathogen exposure⁶⁶⁶⁶⁶⁶⁶⁶. This checks the **specificity of AMP production**.
- **Functional Assay:** Survival is the ultimate measure of the AMP's effectiveness, testing if the AMP can actually kill the pathogen⁷.
- **Pathogens Used:** The experiment involves comparing responses to a **fungus** (*Neurospora crassa*) and a **bacterium** (*Micrococcus luteus*)⁸. (Only the fungus results are shown in the graph).

 Detailed Graph Interpretation (*Neurospora crassa* Infection)

The survival curves demonstrate the protective function (or lack thereof) of the two tested AMPs against the fungus *Neurospora crassa*⁹.

Group	Line Color	Survival Rate at 120 Hours	AMP Specificity Conclusion
Mutant	Blue Line		$\approx 0\%$ 101010 Serves as the negative control; confirms that without the AMPs, the infection is lethal ¹¹ .
Wild Type	Red Line		$\approx 80\%$ 121212 Serves as the positive control; shows the natural, full immune response is highly effective ¹³ .
Mutant + Defensin	Purple Line		$\approx 15\%$ 141414 Shows poor protection . Although slightly better than the Mutant, the survivorship is low, indicating Defensin is not the primary defense against the fungus ¹⁵ .

Mutant + Drosomycin	Green Line		$\approx 70\%$ 161616	Shows high protection , with survival very close to the Wild type ¹⁷ . This demonstrates that Drosomycin is highly effective at neutralizing the fungus.
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Overall Conclusion on Specificity

The results from the *Neurospora crassa* infection strongly suggest that different AMPs are **pathogen-specific**¹⁸.

- **Drosomycin** provides robust defense against the **fungus**¹⁹.
- **Defensin** provides minimal defense against the **fungus**²⁰.

This indicates that the **innate immune system** uses a targeted, specialized response: an invading pathogen triggers the production of a specific AMP that is most effective against that particular class of microbe. (Further results, not shown, would likely confirm that Defensin is specialized to fight the bacteria *Micrococcus luteus* or another bacterial pathogen).

Antimicrobial Peptides (AMPs)

Antimicrobial Peptides (AMPs) are a key component of the **Innate Immune System** found in all animals and plants¹.

- **Definition and Function:** AMPs are short protein fragments² that serve as a rapid, general defense mechanism³³³³. They are made either when proteins are degraded or are encoded by genes⁴.
- **Mechanism of Action:**

- **Membrane Disruption:** Upon infection, AMPs penetrate into the pathogen's phospholipid membrane and disrupt it by interacting with the lipid bilayer⁵. This creates **holes** in the pathogen's membrane, causing the cell to **lyse** (burst)⁶.
- **Internal Disruption:** AMPs can also interact with cellular proteins and nucleic acids inside the pathogen, disrupting essential functions like cell wall synthesis, enzyme activity, or protein folding⁷⁷⁷⁷⁷⁷⁷⁷⁷.
- **Specificity:** AMPs show a degree of **specificity**; for example, the AMP **drosomycin** in *Drosophila* is highly effective against the **fungus** *Neurospora crassa*, while **defensin** is less effective against it⁸.



Green Fluorescent Protein (GFP)

Green Fluorescent Protein (GFP) is a revolutionary tool used in biological research to visualize molecules and processes inside living cells and organisms⁹.

- **Definition:** GFP is a **genetically encoded fluorescent protein**¹⁰.
- **Function:** When exposed to light of a specific wavelength, GFP re-emits light in the green spectrum (it fluoresces). It's used to **fluorescently tag proteins** in cells and animals¹¹.
- **Mechanism in Experiments:** To track the production of a protein like an AMP, scientists can **fuse the AMP gene with the GFP gene**¹².
 - The resulting **AMP::GFP fusion protein** is produced when the cells translate the fused gene¹³.
 - If the cell is producing the AMP, the cell (or tissue) will **fluoresce**, allowing researchers to detect where and when the AMP is being made in response to a pathogen¹⁴¹⁴¹⁴¹⁴.
- **Variants:** Variants of GFP can be made by **mutating the original gene** to produce different colors or optimize properties¹⁵.

